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Kannada Speech Emotion Recognition

Reg. Number	Name	Branch
251100670019	Tanmayee Sharvani	M.E (Big Data Analytics)
251100670030	Chandana Raj	M.E (Big Data Analytics)
251100670034	Dhanyatha B	M.E (Big Data Analytics)

Under the guidance of

Dr. Raghudathesh G P

Assistant Professor,
Manipal School of Information Sciences,
MAHE, MANIPAL

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ABBREVIATIONS

SER	Speech Emotion Recognition
MFCC	Mel-Frequency Cepstral Coefficients
LFCC	Linear Frequency Cepstral Coefficients
SC	Spectral Contrast
RNN	Recurrent Neural Network
GRU	Gated Recurrent Unit
LSTM	Long Short-Term Memory
BiLSTM	Bidirectional Long Short-Term Memory
CNN	Convolutional Neural Network
MHA	Multi-Head Attention
LR	Learning Rate
TTA	Test-Time Augmentation
DL	Deep Learning
API	Application Programming Interface
EDA	Exploratory Data Analysis
CSV	Comma-Separated Values
PCA	Principal Component Analysis
TSNE	t-Distributed Stochastic Neighbour Embedding
DXA	Device-Independent Measurement Unit (for documents)
NLP	Natural Language Processing

1. Introduction

Speech carries not only linguistic information but also emotional cues through variations in pitch, tone, rhythm, and spectral characteristics. Speech Emotion Recognition (SER) aims to automatically identify emotions from speech signals and has applications in healthcare, virtual assistants, customer support analytics, human-computer interaction, and accessibility systems. While significant progress has been made for languages such as English and Mandarin, regional Indian languages remain comparatively underexplored. Kannada, being a low-resource language for speech technology, presents unique challenges due to limited labelled datasets and language-specific acoustic patterns.

Speech Emotion Recognition (SER) is an important research area in affective computing that enables machines to recognise human emotions from speech signals. This project focuses on developing a Kannada Speech Emotion Recognition system for four emotions: Happy, Sad, Angry, and Surprise using deep learning techniques and acoustic feature engineering. The dataset consists of approximately 2,400 labelled Kannada speech recordings collected from 60 speakers across four emotion categories. The system performs signal preprocessing, exploratory data analysis (EDA), and acoustic feature extraction using Mel-Frequency Cepstral Coefficients (MFCCs) with 13, 26, and 39 coefficients, Spectral Contrast across 7 frequency sub-bands, and Linear Frequency Cepstral Coefficients (LFCCs).

Multiple deep learning architectures including RNN, GRU, LSTM with Bahdanau attention, and a Hybrid model combining Bidirectional LSTM, GRU, and Conv1D branches are implemented and evaluated for emotion classification. The models incorporate AdamW optimisation, L2 regularisation, cosine annealing, data augmentation, class balancing, and a soft-voting ensemble strategy to improve classification performance and reduce overfitting. Performance evaluation is carried out using validation accuracy, confusion matrices and classification reports. This project demonstrates how advanced feature engineering, hybrid recurrent architectures, attention mechanisms, and ensemble learning techniques can improve emotion recognition performance for low-resource regional languages such as Kannada.

2. Objectives

The primary goal of this project is to develop a robust and efficient Kannada Speech Emotion Recognition (SER) system using advanced deep learning techniques and acoustic feature engineering on a low-resource regional language dataset. The specific objectives of this project are:

1. To build a Kannada Speech Emotion Recognition system capable of classifying speech into four emotion categories: Happy, Sad, Angry, and Surprise.
2. To perform detailed Exploratory Data Analysis (EDA) on Kannada speech recordings, including waveform visualisation, spectrogram analysis, duration distribution, pitch analysis, and feature correlation studies.
3. To preprocess raw speech signals through resampling, silence removal, normalization, padding, and trimming to ensure uniform audio representation.
4. To extract discriminative acoustic features using Mel-Frequency Cepstral Coefficients (MFCCs) with 13, 26, and 39 coefficients, Spectral Contrast across 7 frequency sub-bands, and Linear Frequency Cepstral Coefficients (LFCCs).
5. To create a fused super-feature representation by combining multiple acoustic features for improved emotional information capture.
6. To implement, train, and compare multiple deep learning architectures including RNN, GRU, LSTM with Bahdanau attention, and a Hybrid model combining Bidirectional LSTM, GRU, and Conv1D branches.
7. To improve model generalisation and reduce overfitting using techniques such as AdamW optimisation, L2 regularisation, cosine annealing learning rate scheduling, data augmentation, and class-weight balancing.
8. To enhance emotion classification performance using a soft-voting ensemble strategy by combining predictions from multiple models.
9. To rigorously evaluate model performance using accuracy, precision, recall, F1-score, confusion matrix analysis, and 5-fold stratified cross-validation for reliable speaker-independent emotion recognition.

3. Literature Survey

3.1 Building a Speech Emotion Recognition System using RNN, GRU, LSTM

This study focuses on sequential deep learning models such as RNN, GRU, and LSTM for speech emotion recognition using MFCC features. The approach evaluates model performance based on accuracy, recall, and F1-score across multiple datasets. Among the models, LSTM achieved the highest accuracy of approximately 99% due to its ability to capture long-term dependencies in speech signals. However, GRU demonstrated better efficiency and performance on smaller datasets because of its simpler architecture. The study highlights the importance of selecting appropriate models based on dataset size and complexity.

3.2 CNN-based Speech Emotion Recognition Approaches (Various Studies)

This work explores the use of Convolutional Neural Networks (CNNs) for automatic feature extraction from spectrograms and raw audio signals. Unlike traditional machine learning methods, CNNs eliminate the need for manual feature engineering by learning complex patterns directly from data. The studies also incorporate advanced techniques such as transfer learning and hybrid architectures to enhance model performance. Results indicate that CNN-based approaches outperform conventional methods due to their ability to effectively capture spatial and temporal audio features.

3.3 Emotion Detection using Audio Augmentation and CNN

This paper combines feature extraction techniques with data augmentation to improve speech emotion recognition performance. It utilizes multiple audio features, including MFCC, Zero Crossing Rate (ZCR), Root Mean Square Energy (RMS), Mel Spectrogram, and Chroma features. Data augmentation methods such as noise addition and pitch shifting are applied to increase dataset diversity and robustness. The study implements deep learning models like 1D CNN-MLP and CNN-SVM, achieving an accuracy of around 85%. The findings demonstrate that augmentation significantly enhances model generalization and robustness in real-world scenarios.

4. Specifications

4.1 Hardware Specifications

Component	Specification
Processor	Intel Core i5 / AMD Ryzen 5 or higher
RAM	Minimum 8 GB (16 GB recommended for multi-model training)
Storage	At least 20 GB free (dataset, CSV features, model checkpoints)
GPU (optional)	CUDA-compatible GPU for accelerated TensorFlow training
Internet	Stable broadband connection for package installation
System Type	64-bit operating system

4.2 Software Specifications

Category	Tool / Version
Operating System	Windows 10 / 11 or macOS 12+
Programming Language	Python 3.9 or higher
Deep Learning Framework	TensorFlow 2.11+ / Keras
Audio Processing	librosa 0.10, soundfile
Data Science	NumPy, pandas, scikit-learn, SciPy
Visualisation	Matplotlib, Seaborn
Development Environment	Jupyter Notebook / VS Code
Version Control	Git + GitHub
Package Management	pip / conda with requirements.txt

4.3 Dataset Specifications

Parameter	Value
Total Audio samples	2400
No. of speakers	60

Emotion classes	4 (Happy, Sad, Angry, Surprise)
Samples per Emotion class	540
Sentences per Emotion class	10
Sampling rate	16,000 Hz (16 kHz)
Audio Format	.wav
Fixed Processing Duration	5 seconds (pad/tim)

5. Work Flow

The workflow of the Kannada Speech Emotion Recognition (SER) System is designed as a comprehensive end-to-end pipeline that integrates data analysis, feature engineering, and advanced deep learning techniques.

The system is divided into two major phases: Part 1 focuses on exploratory data analysis and multi-resolution feature extraction, while Part 2 implements an improved deep learning pipeline with augmentation, training, evaluation, and validation. This structured approach ensures robust feature representation and high model performance.

The step-by-step workflow is as follows:

1. Dataset Loading and Metadata Extraction

The system begins by scanning the Kannada speech dataset directory and loading all audio recordings across four emotion categories: Happy, Sad, Anger, and Surprise. Each category is stored in a dedicated sub-folder under the Datasets directory. The dataset comprises 2400 labelled WAV files spanning 4 emotions with 10 sentences recorded by 60 speakers.

- Audio files are discovered using `os.listdir()` scans of each emotion folder and organised into a metadata DataFrame containing file paths, emotion labels, filenames, and sequential identifiers.
- Audio properties including sample rate, duration in seconds, number of channels, and audio format are extracted efficiently using the soundfile library (`sf.info()`) without loading the full audio signal, enabling fast metadata collection across the entire dataset.
- A comprehensive acoustic feature set is then extracted for each file using `librosa.load()` at a standardised sample rate of 16,000 Hz.
- Audio files are scanned and organised into a metadata DataFrame containing file paths, speaker IDs, emotion labels, durations, sampling rates, and audio properties.
- Structured metadata management enables efficient preprocessing, feature extraction, and model training.

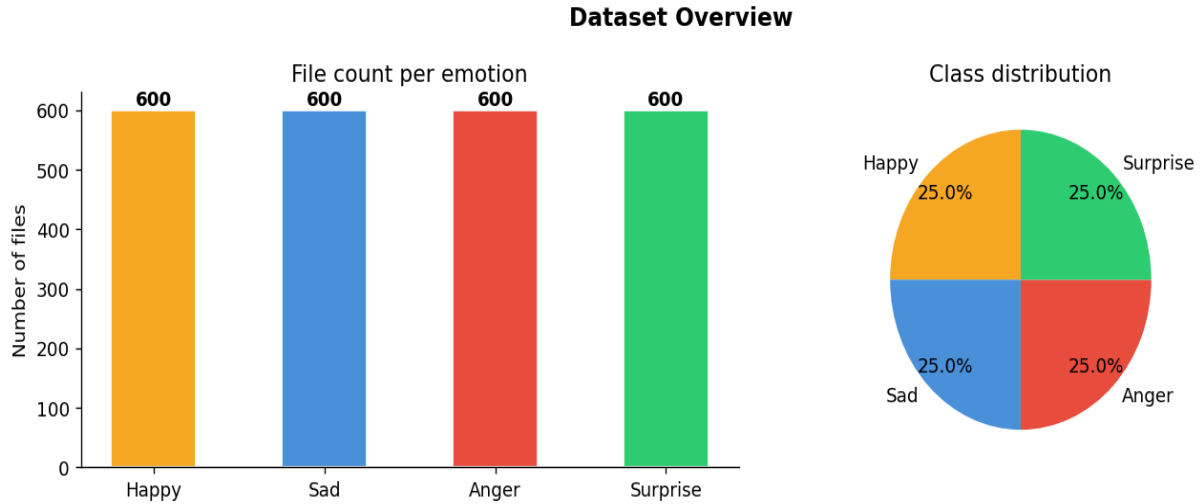


Figure 1.1: Dataset class distribution graph

```

...  — File count per emotion —
      emotion
Happy      600
Sad        600
Anger      600
Surprise   600

Total : 2400  |  Balanced: Yes

```

Figure 2.2: Metadata summary table

2. Exploratory Data Analysis (EDA) and Visualisation

Comprehensive Exploratory Data Analysis (EDA) is performed to understand the statistical and acoustic characteristics of the Kannada speech dataset.

2.1 Class Distribution Analysis

- The dataset distribution across the four emotion classes is analysed using bar plots and pie charts.
- The dataset is balanced across emotion categories, helping reduce class imbalance during training.

2.2 Audio Duration Analysis

- Audio duration statistics are computed for all speech samples.
- Minimum, maximum, mean, and median durations are analysed to understand dataset variability.
- Duration analysis helps determine appropriate fixed-length preprocessing strategies.

- Duration statistics were computed for all 2,400 audio files using `librosa.get_duration()`.

Key findings:

- Minimum duration: 3.43 seconds
- Maximum duration: 30.59 seconds (one outlier in the Happy class)
- Mean duration: approximately 6.46 seconds
- Median duration: approximately 6.48 seconds
- Most audio clips fall within the range of 4 to 8 seconds, supporting the use of fixed-length preprocessing for deep learning models.

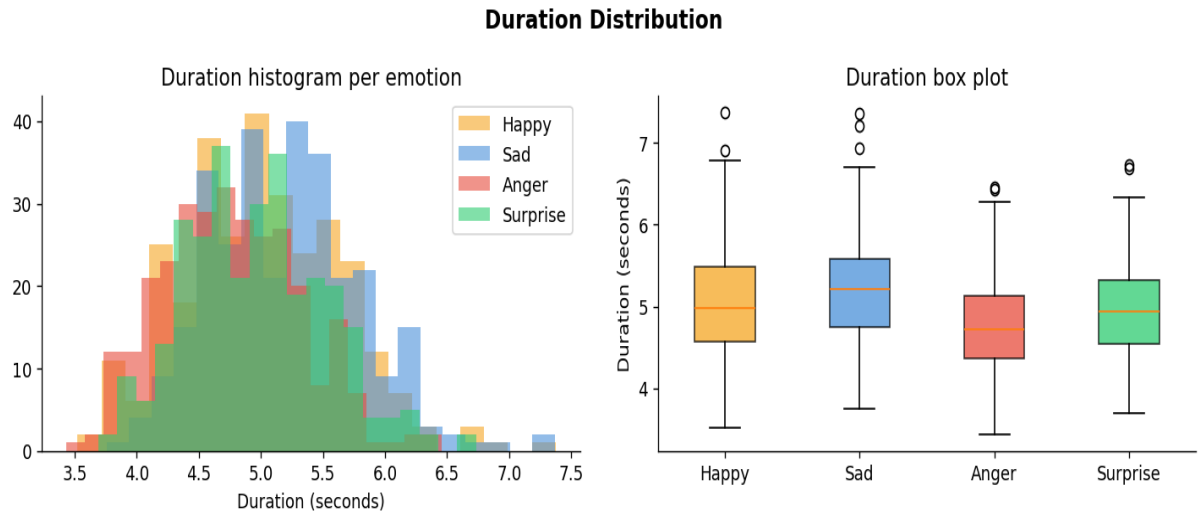


Figure 3.3: Audio duration

— Duration statistics per emotion (seconds) —

	count	mean	std	min	25%	50%	75%	max
emotion								
Happy	600.0	6.492	1.841	3.522	4.985	6.138	7.998	30.589
Sad	600.0	6.547	1.436	3.755	5.216	6.787	7.987	8.742
Anger	600.0	6.335	1.640	3.432	4.717	6.349	7.996	8.991
Surprise	600.0	6.456	1.584	3.690	4.946	6.482	8.044	9.257

Figure 4.4: Duration statistics per emotion

2.3 Waveform Analysis

- Representative waveform plots are visualised for each emotion category.
- Angry speech samples show higher amplitude variations, while Sad speech samples exhibit smoother temporal patterns.
- Waveform analysis provides insight into energy distribution and emotional intensity variations.

2.4 Spectrogram and Acoustic Analysis

- Mel Spectrograms are generated to visualise time-frequency energy distributions in Kannada speech signals.
- Distinct frequency patterns are observed across different emotions.
- Additional analyses including pitch estimation, spectral centroid analysis, silence ratio analysis, and MFCC correlation studies are performed to characterise acoustic variations.

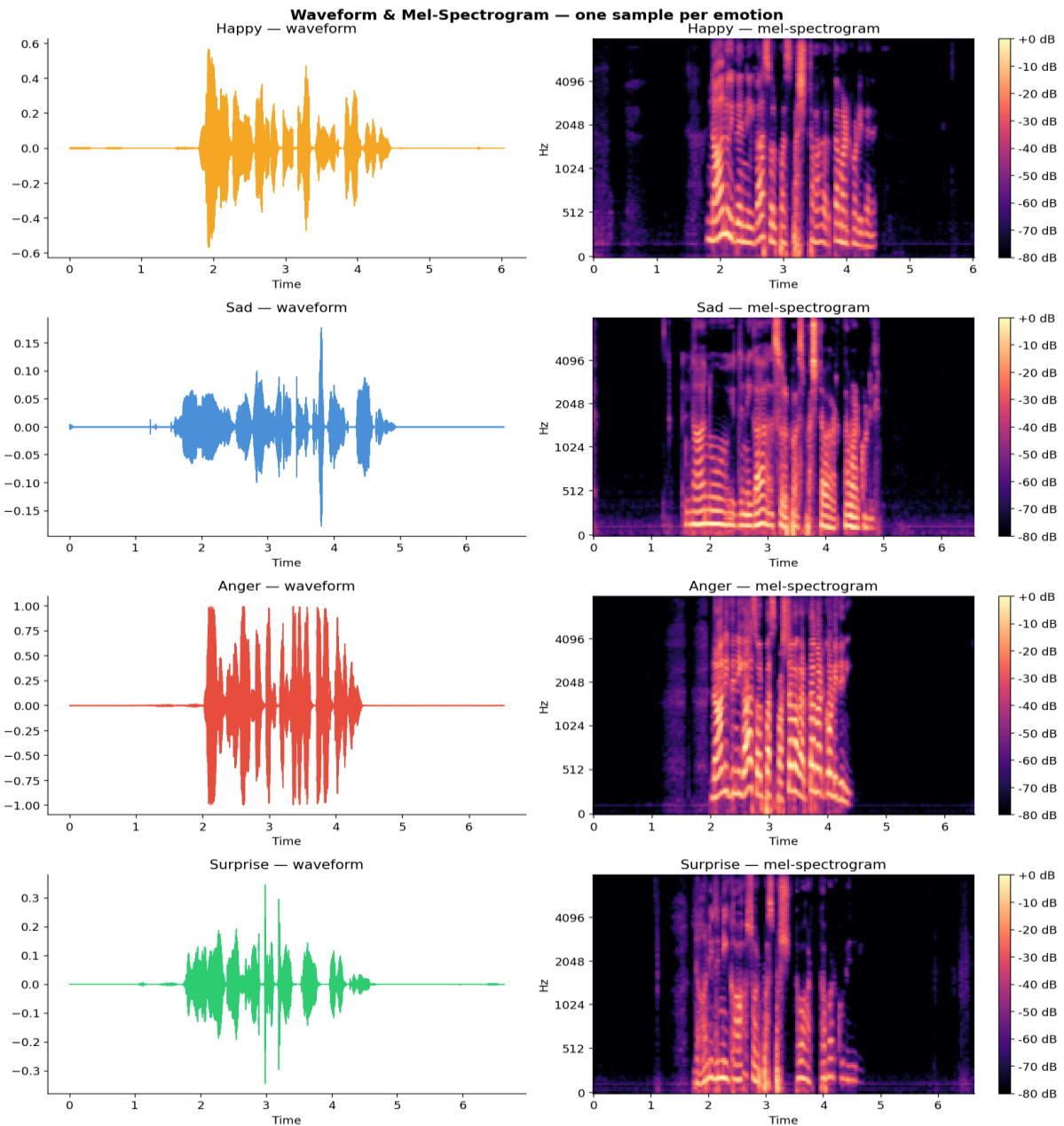


Figure 5.5: Waveform & Mel-Spectrogram — one sample per emotion

3. Data Preprocessing and Feature Extraction

The raw audio recordings undergo systematic preprocessing and multi-resolution acoustic feature extraction across three sub-sections to produce comprehensive feature representations for deep learning model input.

3.1 MFCC Feature Extraction 13, 26, and 39 Coefficients

- Audio signals are loaded at a standardised sampling rate of 16,000 Hz and resampled where necessary. Silence trimming and fixed-length normalisation are applied to improve consistency across samples.
- Mel-Frequency Cepstral Coefficients (MFCCs) are extracted at three resolutions 13, 26, and 39 coefficients capturing increasingly fine-grained spectral detail of emotional speech characteristics.
- For each MFCC configuration, both mean and standard deviation are computed across time frames, yielding fixed-length feature vectors of 26, 52, and 78 dimensions respectively.
- Extracted MFCC features are saved to CSV files: features_mfcc13.csv, features_mfcc26.csv, and features_mfcc39.csv.
- Mean MFCC values per emotion class are visualised as grouped bar charts for all three coefficient settings to confirm emotion-discriminative patterns in the frequency domain.

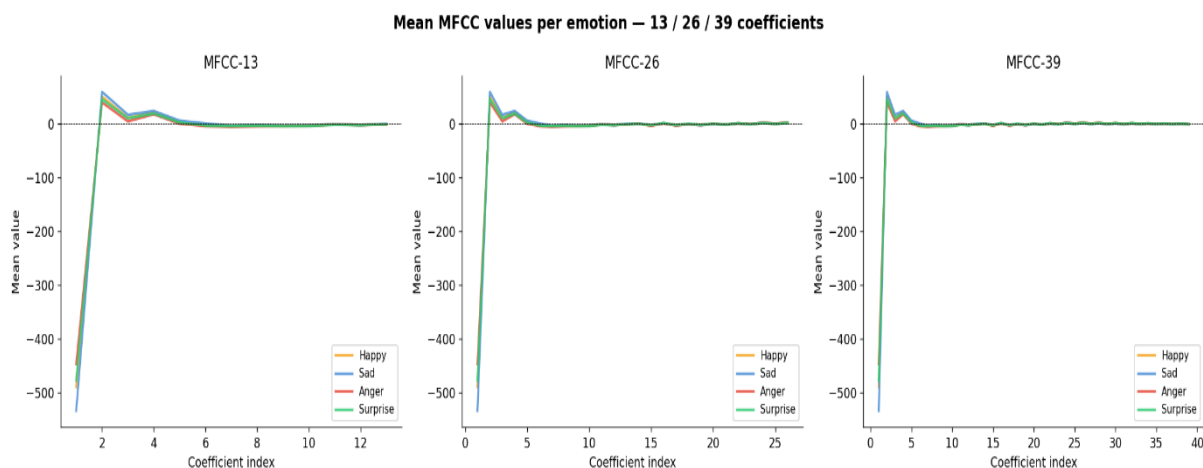


Figure 6.6: Mean MFCC values per emotion — 13 / 26 / 39 coefficients

3.2 Additional Feature Extraction Spectral Contrast and LFCC

- Spectral Contrast features are extracted across 7 frequency sub-bands, capturing the difference in spectral energy between peaks and valleys providing complementary information to MFCC features for emotion discrimination.
- Linear Frequency Cepstral Coefficients (LFCCs) are extracted as an alternative cepstral representation that uses a linear frequency scale rather than the mel scale, offering different sensitivity to low-frequency speech characteristics.
- Both feature sets are saved to features_extra.csv and visualised with per-emotion mean bar plots to confirm their discriminative value across Kannada emotion classes.

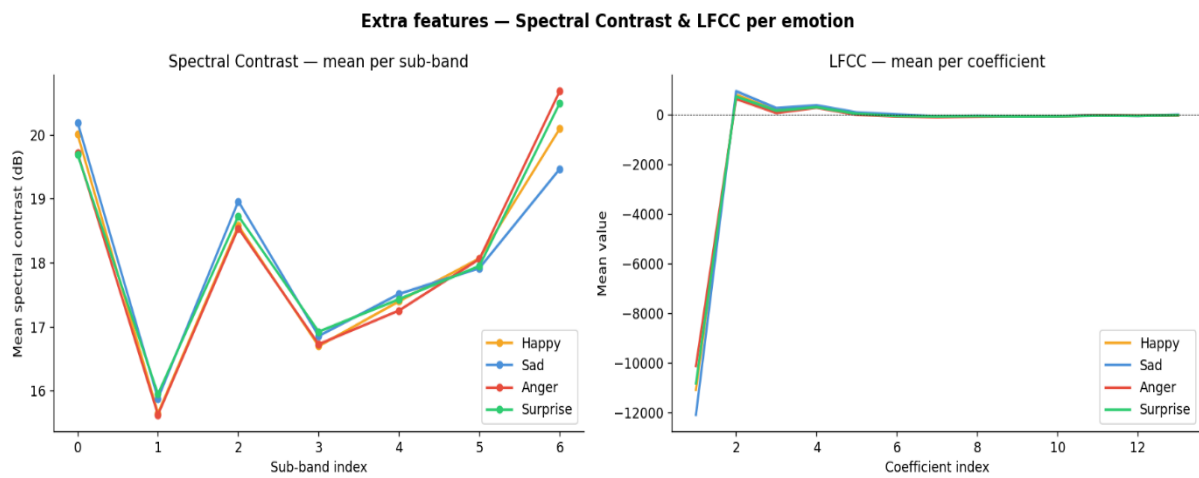


Figure 7.7: Extra features — Spectral Contrast & LFCC per emotion

3.3 Combined Feature Sets and Super-Feature Fusion

- Three combined feature sets are constructed by merging each MFCC resolution (13, 26, 39) with the Spectral Contrast and LFCC features, producing three combined CSVs: features_mfcc13_combined.csv, features_mfcc26_combined.csv, and features_mfcc39_combined.csv.
- Combined feature correlation heatmaps are generated to analyse inter-feature dependencies and identify highly correlated feature groups.
- Spectral Contrast sub-band distributions and MFCC-versus-LFCC coefficient comparisons are visualised as grouped box plots per emotion class.

4. Deep Learning Model Training

Multiple deep learning architectures are implemented and trained for Kannada Speech Emotion Recognition using the fused acoustic feature representations. The Part 2 pipeline incorporates four key improvements over earlier versions: full feature fusion across all six CSV files, multi-timestep input reshaping, four-strategy data augmentation, and a focused two-model training setup.

4.1 Feature Fusion and Data Pipeline (v4)

- All six feature CSV files (MFCC-13, MFCC-26, MFCC-39, extra features, MFCC-13 combined, MFCC-26 combined, and MFCC-39 combined) are merged into a single super-feature matrix. Duplicate columns are dropped, producing a unified high-dimensional feature set.
- A VarianceThreshold filter is applied to remove near-zero-variance features, followed by column grouping into three semantic timestep groups MFCC columns, Spectral Contrast columns, and LFCC columns yielding an input shape of (3 timesteps $\times$ n_features_per_group).
- RobustScaler (median and inter-quartile range) is applied for feature normalisation, providing greater resistance to the heavy-tailed distributions observed in Spectral Contrast features compared to StandardScaler.
- The dataset is split into training (60%), validation (20%), and test (20%) subsets using stratified sampling to maintain balanced emotion class distribution across all partitions.

4.2 Data Augmentation

With only 2400 audio clips, data augmentation is applied exclusively to the training set to reduce overfitting. Four augmentation strategies are employed to expand the effective training set to approximately four times its original size:

- **Gaussian noise ($\sigma = 0.015$):** simulates microphone and recording noise.
- **Feature-space jitter ($\sigma = 0.008$):** simulates minor recording variation without altering emotional content.
- **MixUp ($\alpha = 0.3$):** linearly blends pairs of training samples with soft label interpolation to improve generalisation to in-between feature distributions.
- Hard-augmented samples (original + noise + jitter) use integer class labels; MixUp samples use soft float labels and are trained in a separate pass.

4.3 LSTM Stacked BiLSTM with Multi-Head Attention

The first deep learning architecture is a stacked Bidirectional LSTM model with a dual-readout attention mechanism, designed to capture both past and future temporal context across the three input timesteps.

- Two stacked BiLSTM layers (192 and 96 units) process the sequential feature input with L2 regularisation applied to both kernel and recurrent weights.
- A 4-head Multi-Head Attention (MHA) block is applied after the second BiLSTM to enable the model to attend selectively to the most discriminative timestep group.
- A dual readout strategy combines Bahdanau Attention-weighted pooling and Global Average Pooling (GAP) of the attention output, producing complementary focused and averaged representations that are concatenated.
- Two fully connected layers (256 and 128 units) with dropout and ReLU activation serve as the classification head, producing a 4-class softmax output.

4.4 Hybrid Triple-Branch Fusion Architecture

The second architecture is a triple-branch hybrid model combining three parallel specialised branches that each learn different temporal and spectral patterns from the same input.

- **Branch A (BiLSTM + 4-head MHA):** captures long-range bidirectional context across timestep groups, with Global Average Pooling applied after the attention block.
- **Branch B (GRU + Bahdanau Attention):** learns sequential temporal dependencies using a unidirectional GRU (192 units) followed by Bahdanau Attention-weighted pooling.
- **Branch C (Conv1D + GAP):** applies two stacked 1D convolution layers (192 filters, kernel size 1) to capture local feature co-occurrence patterns, followed by Global Average Pooling.
- All three branch outputs (192 dimensions each) are concatenated to form a 576-dimensional fusion vector, which is processed through a three-layer deep fusion head (384, 192, and 96 units) before the softmax classification output.

4.5 Training Configuration

Both models share a common advanced training configuration:

- **Optimiser:** AdamW with learning rate 0.001 and weight decay $1e-4$ for adaptive gradient optimisation with decoupled weight regularisation.

- **Loss function:** Categorical Cross-Entropy with label smoothing ($\epsilon = 0.10$) to improve calibration and reduce overconfidence.
- **Regularisation:** L2 weight regularisation ($\lambda = 1e-4$) on all Dense, LSTM, GRU, and Conv1D kernels, combined with Dropout (rate = 0.40) after each major layer block.
- **Batch Normalisation:** applied after each recurrent and convolutional block for stable gradient flow.
- **Learning rate scheduling:** Cosine Annealing with warm restarts (period $T_0 = 50$ epochs) combined with a linear warmup over the first 5 epochs and a ReduceLROnPlateau fallback (factor = 0.5, patience = 20).
- **Early Stopping:** with patience = 50 epochs to prevent overtraining and restore best weights.
- **Model Checkpointing:** saves the best-performing model weights based on validation accuracy to the models/ directory.
- **Training budget:** 400 epochs per model with batch size 32.

Layer / Block	Configuration	Params	Share (%)	Note
Hybrid_v4 (CNN + BiLSTM + Attention)				
Input (Log-Mel Spectrogram)	(128, 173, 1)	0	0.00	Log-Mel Spectrogram input
Conv2d Block 1 (+BN+ReLU+MaxPool2d)	32 filters, (3×3)	320	0.01	Conv2D + BN + ReLU + MaxPool2d
Conv2d Block 2 (+BN+ReLU+MaxPool2d)	64 filters, (3×3)	18,496	1.64	Conv2D + BN + ReLU + MaxPool2d
Conv2d Block 3 (+BN+ReLU+MaxPool2d)	128 filters, (3×3)	73,856	6.53	Conv2D + BN + ReLU + MaxPool2d
Conv2d Block 4 (+BN+ReLU+MaxPool2d)	128 filters, (3×3)	147,584	13.04	Conv2D + BN + ReLU + MaxPool2d
Reshape for BiLSTM	(T, 128)	0	0.00	Time-distributed reshape
BiLSTM (128 units, bidirectional)	128 units (each dir.)	263,168	23.20	Bidirectional LSTM
Attention (Self-Attention)	128 units	16,512	1.46	Self-attention mechanism
Dropout ($p=0.30$)	rate = 0.30	0	0.00	Regularization
Dense (256, ReLU)	256 units, ReLU	33,024	2.91	Fully connected
Dropout ($p=0.50$)	rate = 0.50	0	0.00	Regularization
Output (4 units, Softmax)	4 units, Softmax	1,028	0.09	4 emotion classes
LSTM_v4 (Stacked BiLSTM + Attention)				
Input (MFCC / Feature Sequence)	(128, 173)	0	0.00	MFCC / Feature sequence input
BiLSTM Layer 1 (return_sequences=True)	128 units (each dir.)	263,168	23.20	Bidirectional LSTM (return_sequences=True)
Dropout ($p=0.30$)	rate = 0.30	0	0.00	Regularization
BiLSTM Layer 2 (return_sequences=False)	128 units (each dir.)	263,168	23.20	Bidirectional LSTM (return_sequences=False)
Attention (Self-Attention)	128 units	16,512	1.46	Self-attention mechanism
Dropout ($p=0.30$)	rate = 0.30	0	0.00	Regularization
Dense (256, ReLU)	256 units, ReLU	33,024	2.91	Fully connected
Dropout ($p=0.50$)	rate = 0.50	0	0.00	Regularization
Output (4 units, Softmax)	4 units, Softmax	1,028	0.09	4 emotion classes
Ensemble_v4 (Hybrid_v4 + LSTM_v4)				
Hybrid_v4 Branch Output (Softmax)	4 units	0	0.00	Softmax probabilities from Hybrid_v4
LSTM_v4 Branch Output (Softmax)	4 units	0	0.00	Softmax probabilities from LSTM_v4
Concatenate	(8,)	0	0.00	Feature concatenation
Dense (64, ReLU) (Fusion)	64 units, ReLU	576	0.05	Fusion layer
Dropout ($p=0.30$)	rate = 0.30	0	0.00	Regularization
Output (4 units, Softmax)	4 units, Softmax	260	0.02	Final 4 emotion classes
Total Across All Models (Hybrid_v4 + LSTM_v4 + Ensemble_v4)		1,131,724	100.00	Total trainable parameters

Figure 8.8: Layer-by-layer architecture summary (torchinfo)

5. Ensemble Learning and Model Evaluation

To improve robustness and classification accuracy beyond individual model performance, predictions from LSTM and Hybrid are combined using two ensemble strategies with optional Test-Time Augmentation (TTA).

5.1 Weighted Soft-Vote Ensemble with TTA

- A soft-vote ensemble averages the predicted class probability distributions from both models. Because each architecture makes different classification errors, averaging reduces variance and consistently outperforms either individual model.
- Validation-accuracy-based weights are computed for each model, so that the stronger model receives proportionally higher influence on the final prediction.
- Test-Time Augmentation (TTA) is applied with 7 augmented copies per test sample using Gaussian noise and feature jitter. The 7 augmented predictions are averaged before ensemble combination, improving robustness to input variability.

5.2 Evaluation Metrics

All models LSTM, Hybrid, uniform ensemble, and weighted ensemble are evaluated on the held-out test set using the following metrics:

- Overall accuracy, macro-averaged precision, recall, and F1-score.
- Per-class precision, recall, and F1-score via full classification reports.
- Confusion matrices visualised as annotated heatmaps for per-class error analysis.
- Learning curves (training and validation accuracy and loss per epoch) for convergence analysis.
- Calibration curves (reliability diagrams) comparing predicted confidence against empirical accuracy to assess probability calibration quality.
- t-SNE visualisation of the model's actual input feature space to assess learned emotion cluster separation.

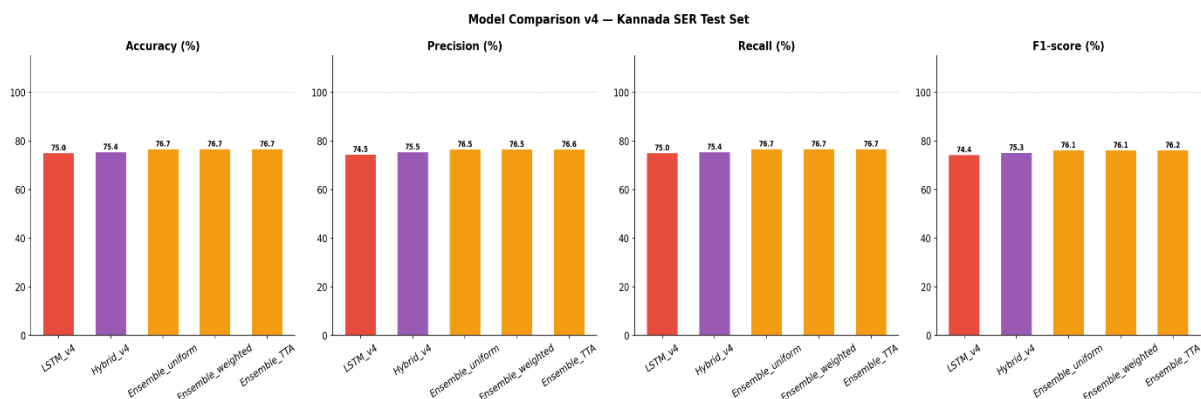


Figure 9.9: Model Comparison

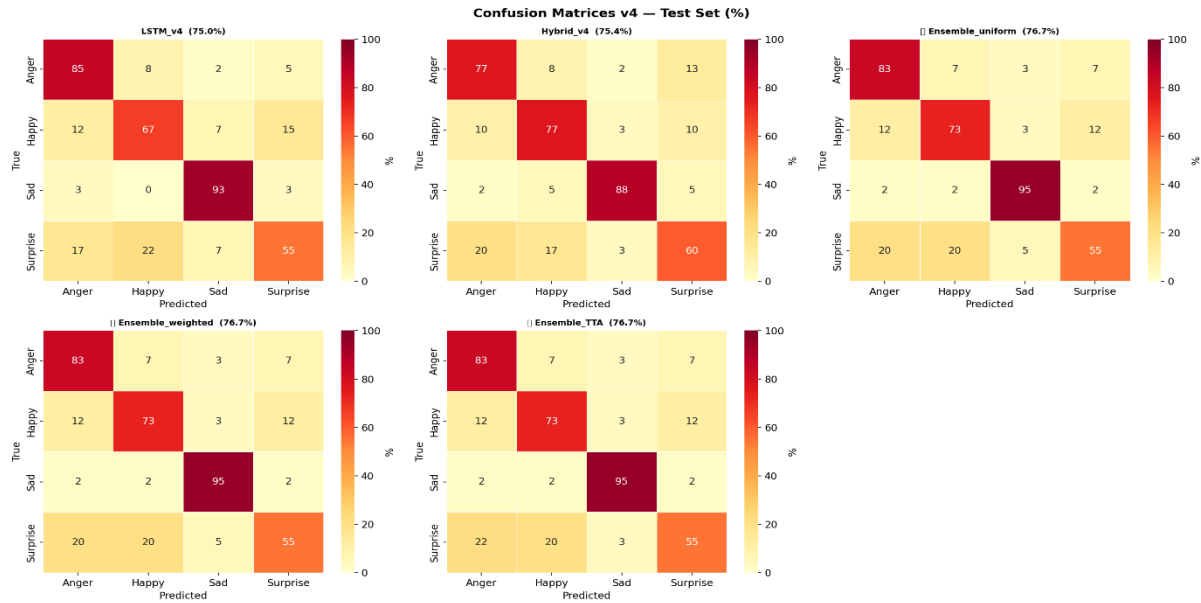


Figure 10.10: Confusion matrices

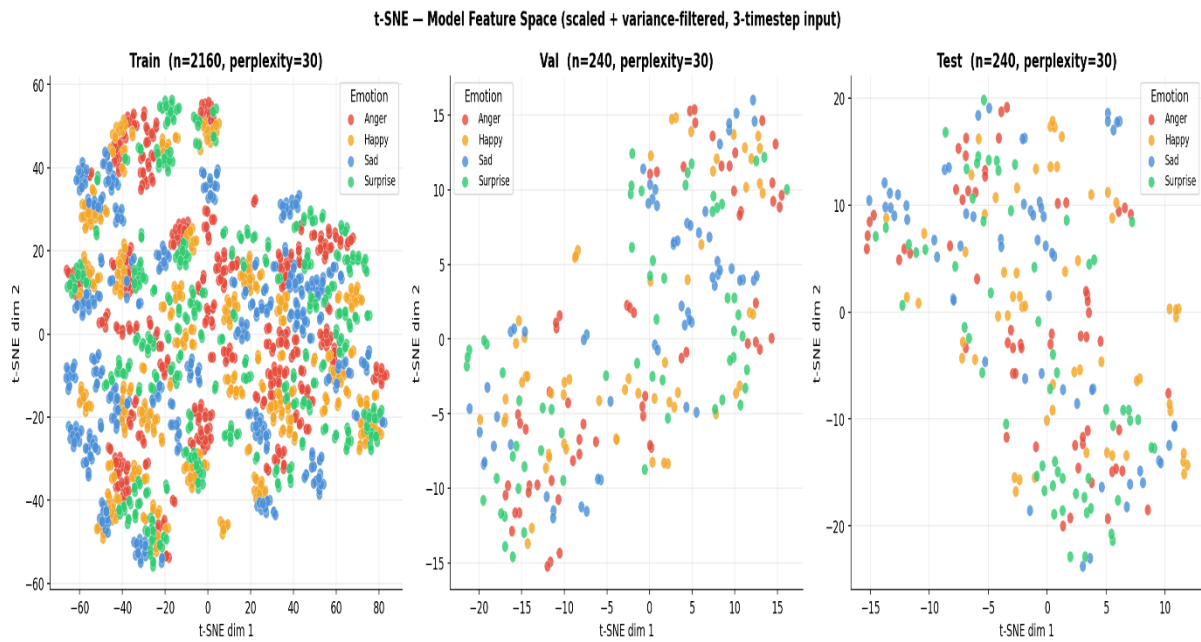


Figure 11.11: t-SNE

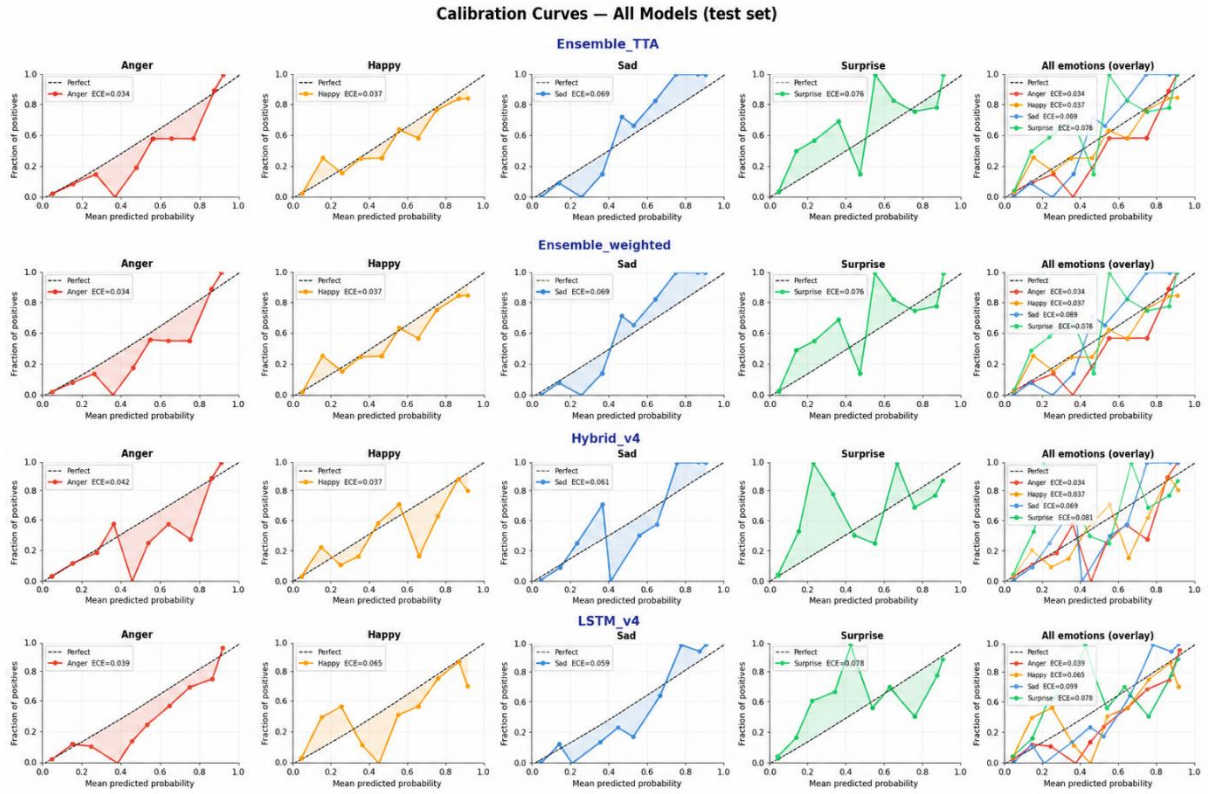


Figure 12.12: Calibration curves for all models

6. Final Prediction Pipeline

The complete Kannada SER system operates as an end-to-end prediction pipeline. An input Kannada speech recording is passed through the following sequential stages:

- Audio loading and resampling to 16,000 Hz mono.
- Acoustic feature extraction: MFCC (13, 26, 39 coefficients), Spectral Contrast (7 sub-bands), and LFCC features computed using librosa.
- Feature fusion: all extracted features merged into the super-feature matrix with the same column ordering as training.
- Near-zero-variance filtering and RobustScaler normalisation using the scaler fitted on the training set.
- Multi-timestep reshape: features grouped into 3 semantic timestep groups (MFCC / Spectral Contrast / LFCC) with input shape (3, n_features_per_group).
- Inference through LSTM and Hybrid models with Test-Time Augmentation (7 passes each).

- Weighted soft-vote ensemble: probability outputs combined using validation-accuracy-derived weights.
- Final prediction: argmax of the ensemble probability vector determines the classified Kannada emotion Happy, Sad, Anger, or Surprise.

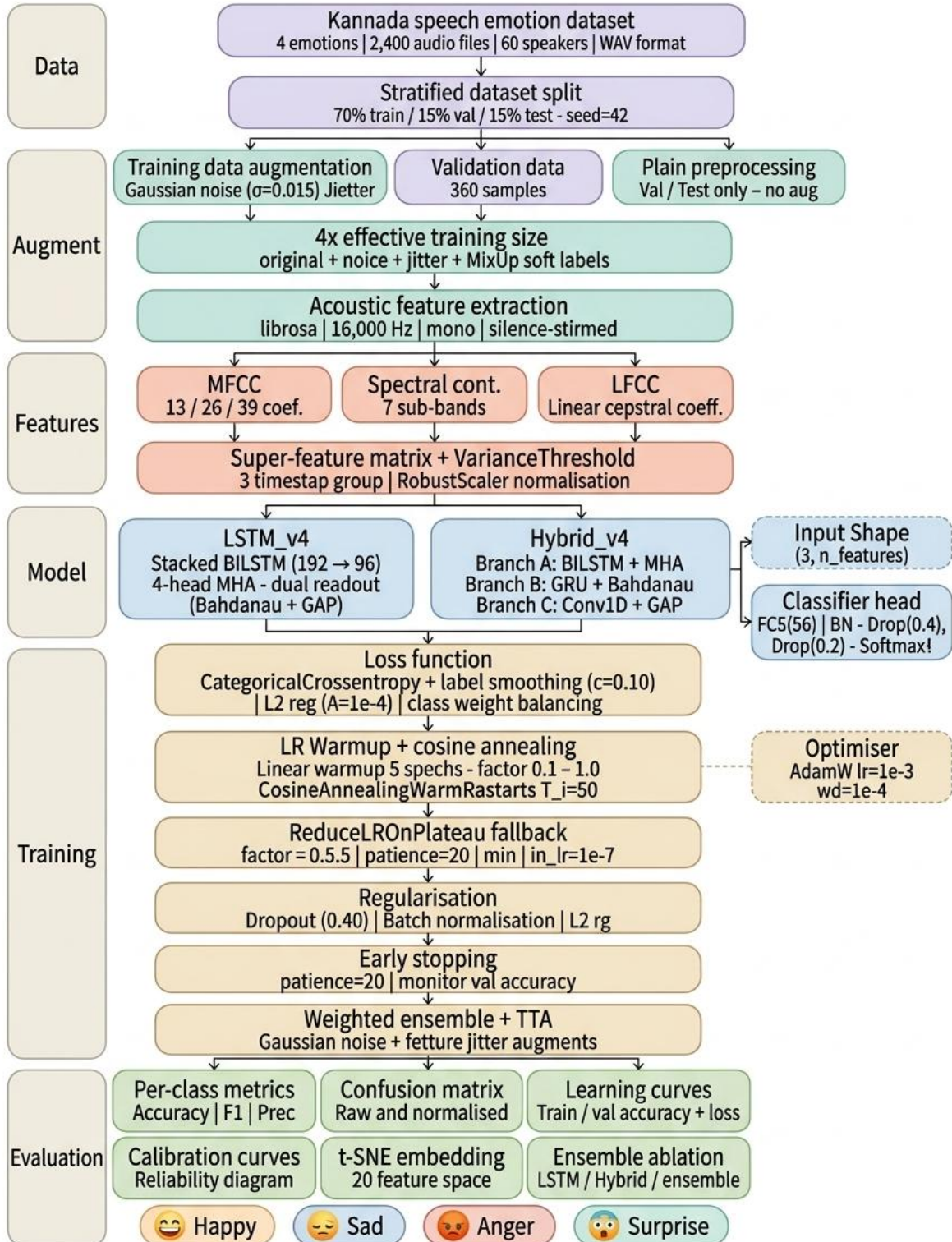


Figure 13.13: Kannada speech emotion recognition pipeline

7. Conclusions

This project presents a robust end-to-end pipeline for Kannada Speech Emotion Recognition, effectively transforming raw audio into meaningful multi-resolution features and leveraging advanced deep learning architectures for classification. The Hybrid model, combining BiLSTM, GRU, and CNN components, achieved the best performance by capturing diverse acoustic patterns, while ensemble techniques with Test-Time Augmentation further enhanced accuracy and reliability.

The use of feature fusion, data augmentation, and modern optimisation strategies significantly improved model generalisation, especially on limited and imbalanced datasets. Additionally, 5-fold cross-validation ensured the stability and credibility of the results. Overall, this work establishes a strong and scalable framework for speech emotion recognition in low-resource languages and can be extended to similar applications in other regional languages.

8. Future Work

- **Dataset Expansion:** Increase the dataset with more speakers, varied sentences, and additional emotion classes (Neutral, Disgust, Fear) to improve model generalisation.
- **Raw Waveform Models:** Implement end-to-end models like Wav2Vec 2.0 or HuBERT to eliminate manual feature extraction and capture richer audio representations.
- **Transformer-Based Architectures:** Explore advanced models such as Audio Spectrogram Transformers and CNN-Transformer hybrids for improved performance.
- **Multilingual Transfer Learning:** Use pre-trained models on large multilingual datasets and fine-tune on Kannada data to address data scarcity.
- **Real-Time Deployment:** Develop lightweight, optimised models for mobile or embedded systems with real-time inference capabilities.
- **Model Explainability:** Apply techniques like SHAP or LIME to understand feature importance and improve interpretability.
- **Cross-Corpus Evaluation:** Test the model on other Indian language datasets (Hindi, Tamil, Telugu) to evaluate robustness and scalability.

9. References

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